

2(a) Books rated above 4 — \$gt

```
bookFlowDB> db.book_metadata.find(
|   { avgRating: { $gt: 4 } },
|   { title: 1, avgRating: 1, _id: 0 }
| )
[
  { title: 'The Silent Algorithm', avgRating: 4.6 },
  { title: 'The Age of Code', avgRating: 4.2 }
]
bookFlowDB> █
```

2(b) Books available in specific formats — \$in

```

}
bookFlowDB> db.book_metadata.find(
|   { avgRating: { $gt: 4 } },
|   { title: 1, avgRating: 1, _id: 0 }
| )
[
  { title: 'The Silent Algorithm', avgRating: 4.6 },
  { title: 'The Age of Code', avgRating: 4.2 }
]
bookFlowDB> db.book_metadata.find(
|   { formats: { $in: ["Digital", "Audiobook"] } },
|   { title: 1, formats: 1, _id: 0 }
| )
[
  {
    title: 'The Silent Algorithm',
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ]
  },
  { title: 'The Age of Code', formats: [ 'Digital', 'Paperback' ] }
]
```

2(c) Combine both with \$and (implicit)

```
bookFlowDB> db.book_metadata.find({
|   genre: "Technology",
|   avgRating: { $gt: 4 }
| }).pretty()
[
  {
    _id: ObjectId('6aad531850578afdcce82a09'),
    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      },
      {
        member_id: 'M1042',
        rating: 4,
        comment: 'Good but the Digital edition had formatting issues.',
        date: ISODate('2023-02-20T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6aad534550578afdcce82a0a'),
    title: 'The Age of Code',
    author: 'F. Fernandes',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [ 'Digital', 'Paperback' ],
    specs: { pages: 216, fileSize_MB: 2.1, drm: false },
    avgRating: 4.2,
    reviews: [
      {
        member_id: 'M10281',
        rating: 4,
        comment: 'Solid Digital read, clean layout.',
        date: ISODate('2022-05-10T00:00:00.000Z')
      },
      {
        member_id: 'M10202',
```

2(d) Regex query – books with "Digital" format

```
]
bookFlowDB> db.book_metadata.find(
|   { formats: { $regex: "Digital", $options: "i" } },
|   { title: 1, formats: 1 }
[| )
[
  {
    _id: ObjectId('6aad531850578afdcce82a09'),
    title: 'The Silent Algorithm',
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ]
  },
  {
    _id: ObjectId('6aad534550578afdcce82a0a'),
    title: 'The Age of Code',
    formats: [ 'Digital', 'Paperback' ]
  }
]
bookFlowDB> █
```

2(e) Regex query – reviews mentioning "Digital" in comments

```
]
bookFlowDB> db.book_metadata.find(
|   { "reviews.comment": { $regex: "Digital", $options: "i" } },
|   { title: 1, "reviews.$": 1 }
[| )
[
  {
    _id: ObjectId('6aad531850578afdcce82a09'),
    title: 'The Silent Algorithm',
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6aad534550578afdcce82a0a'),
    title: 'The Age of Code',
    reviews: [
      {
        member_id: 'M10281',
        rating: 4,
        comment: 'Solid Digital read, clean layout.',
        date: ISODate('2022-05-10T00:00:00.000Z')
      }
    ]
  }
]
bookFlowDB> █
```

Question 1 – Document Modelling.

```

bookFlowDB> db.book_metadata.insertMany([
  {
    title: "The Silent Algorithm",
    author: "R. Menon",
    genre: "Technology",
    publishedYear: 2022,
    formats: ["Hardcover", "Digital", "Audiobook"],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: "K. Rao",
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: "M1001",
        rating: 5,
        comment: "Excellent deep dive into digital systems design.",
        date: ISODate("2023-01-15")
      },
      {
        member_id: "M1042",
        rating: 4,
        comment: "Good but the Digital edition had formatting issues.",
        date: ISODate("2023-02-20")
      }
    ]
  },
  {
    title: "Ocean of Code",
    author: "A. Fernandes",
    genre: "Technology",
    publishedYear: 2021,
    formats: ["Digital", "Paperback"],
    specs: {
      pages: 210,
      fileSize_MB: 2.1,
      drm: false
    },
    avgRating: 4.2,
    reviews: [
      {
        member_id: "M2001",
        rating: 4,
        comment: "Solid Digital read, clean layout.",
        date: ISODate("2022-05-10")
      },
      {
        member_id: "M2002",
        rating: 5,
        comment: "Best programming book this year.",
        date: ISODate("2022-06-01")
      }
    ]
  }
])
bookFlowDB> db.book_metadata.insertMany([
  {
    title: "The Silent Algorithm",
    author: "R. Menon",
    genre: "Technology",
    publishedYear: 2022,

```

```

genre: "Technology",
publishedYear: 2022,
formats: ["Hardcover", "Digital", "Audiobook"],
specs: {
  pages: 340,
  fileSize_MB: 4.2,
  drm: true,
  narrator: "K. Rao",
  runtime_min: 610
},
avgRating: 4.6,
reviews: [
  {
    member_id: "M1001",
    rating: 5,
    comment: "Excellent deep dive into digital systems design.",
    date: ISODate("2023-01-15")
  },
  {
    member_id: "M1042",
    rating: 4,
    comment: "Good but the Digital edition had formatting issues.",
    date: ISODate("2023-02-20")
  }
]
},
{
  title: "Ocean of Code",
  author: "A. Fernandes",
  genre: "Technology",
  publishedYear: 2021,
  formats: ["Digital", "Paperback"],
  specs: {
    pages: 210,
    fileSize_MB: 2.1,
    drm: false
  },
  avgRating: 4.2,
  reviews: [
    {
      member_id: "M2001",
      rating: 4,
      comment: "Solid Digital read, clean layout.",
      date: ISODate("2022-05-10")
    },
    {
      member_id: "M2002",
      rating: 5,
      comment: "Best programming book this year.",
      date: ISODate("2022-06-01")
    }
  ]
},
{
  title: "Whispers of Yesterday",
  author: "S. Kapoor",
  genre: "Fiction",
  publishedYear: 2019,

```

```

    title: "Whispers of Yesterday",
    author: "S. Kapoor",
    genre: "Fiction",
    publishedYear: 2019,
    formats: ["Hardcover", "Paperback"],
    specs: {
      pages: 410,
      binding: "Sewn"
    },
    avgRating: 3.8,
    reviews: [
      {
        member_id: "M3001",
        rating: 3,
        comment: "Slow start but strong ending.",
        date: ISODate("2020-01-15")
      },
      {
        member_id: "M3002",
        rating: 4,
        comment: "Beautifully written prose.",
        date: ISODate("2020-02-11")
      }
    ]
  }
]
})
}
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6aad5d3050578afdcce82a11'),
    '1': ObjectId('6aad5d3050578afdcce82a12'),
    '2': ObjectId('6aad5d3050578afdcce82a13')
  }
}
}
bookFlowDB> db.book_metadata.countDocuments()
3
bookFlowDB> █

```